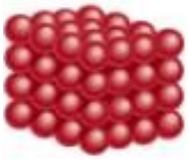
Cut out the labels, diagrams and descriptions below, then arrange them on your page to explain how substances change state. You will need to draw in most of your own arrows. Glue your pieces in place.

Solid **Liquid**

Gas







Melting

Sublimation

Evaporation

Sublimation

Freezing

Condensation

(solidification)

(vaporisation)

Boiling

Energy Increasing

Energy Decreasing

Where a solid turns directly into a gas without going through a liquid phase

As heat adds energy to the solid, the particles on the edge start to vibrate so violently that they break away from the others.

The bonds holding the particles together in a liquid are just strong enough to hold them together. When heat is added and the particles start vibrating more violently the bonds can no longer hold the particles together

Some gases change directly into solids without passing through a liquid state.

Here the particles in a liquid lose so much heat they slow down and turn into a solid.

As gas loses heat energy the particles slow down and attract to each other forming bonds. These bonds hold the particles together in the form of a liquid

A special case where evaporation occurs throughout the liquid forming bubbles. Only happens at the temperature called the Boiling *Point.*